

ORIGINAL RESEARCH

Real-time passenger flow anomaly detection in metro system

Xiulan Wei¹  | Yong Zhang² | Xinyu Zhang²  | Qibin Ge³ | Baocai Yin²

¹Beijing Key Laboratory of Multimedia and Intelligent Software Technology, Beijing Institute of Artificial Intelligence, College of Metropolitan Transportation, Beijing University of Technology, Beijing, China

²Beijing Key Laboratory of Multimedia and Intelligent Software Technology, Beijing Institute of Artificial Intelligence, the Faculty of Information Technology, Beijing University of Technology, Beijing, China

³Scientific research management department, Beijing Intelligent Transportation Development Center, Beijing, China

Correspondence

Yong Zhang, Beijing Key Laboratory of Multimedia and Intelligent Software Technology, Beijing Institute of Artificial Intelligence, the Faculty of Information Technology, Beijing University of Technology, Beijing 100124, China.
Email: zhangyong2010@bjut.edu.cn

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Abstract

Real-time passenger-flow anomaly detection at all metro stations is a very critical task for advanced Internet management. Robust principal component analysis (RPCA) based method has often been employed for anomaly detection task of multivariate time series data. However, it ignores the spatio-temporal features of regular passenger-flow patterns, resulting in a decrease in the accuracy of anomaly detection. In this paper, RT-STRPCA model integrating temporal periodicity and spatial similarity is proposed to address the above issues. RT-STRPCA model detects anomalies by decomposing the observation data into normal component and anomaly component. The spatio-temporal constraints are taken into account to extract anomalies more accurately. The real-time anomaly detection are realized by a sliding window. The performance of RT-STRPCA model is evaluated on synthetic datasets and real-world datasets. The experimental results on synthetic datasets demonstrate that the method achieves more accurate real-time anomaly detection than baseline approaches and the result on real-world datasets verify the utility and effectiveness of the proposed method.

1 | INTRODUCTION

Metro system plays an important role in public transportation for its high efficiency and punctuality. The anomaly flow in the metro system, especially the sudden increase in passenger flow, may cause loss of life or property if it is not handled in time. The anomaly passenger-flow detection task in stations is very important for the intelligent operation of metro systems. Detecting anomalies in the early stage helps to take timely intervention measures to reduce the potentially negative impact caused by anomalies. In recent years, large amounts of passenger flow data have been generated and collected, which brings new methods and perspectives for anomaly detection.

Under normal operating conditions, the in-flow or out-flow of each metro station always repeats a similar pattern according to a certain period (e.g. 1 day or 1 week). Unusual events (such as controls, protests, sport events, celebrations, disasters

and other events) can disturb the normal operation of the metro network, which can be reflected by significant fluctuations in passenger flow data. Therefore, we broadly define abnormal passenger flows as data to deviate from normal data patterns in this paper. These abnormal changes of passenger flow data, including the significantly increase or decrease, can be detected from the passenger flow data of the metro system. Thus, our aim is to efficiently find these significant changes in passenger-flow data, which are most likely caused by abnormal events. Given the characteristics of abnormal passenger flows in metro system, it faces the following critical challenge:

- Passenger flow anomalies can happen anywhere in the metro system at any time and most anomalous events are unexpected.
- Abnormal events rarely occur. Moreover, few of them are recorded.

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These challenges make it difficult to directly detect the occurrence of abnormal events based on supervised method. Thus, many unsupervised anomaly detection algorithms have been developed for univariate time series. However, for a complex real-world metro system, there are strong spatial correlations between different stations. Thus, single univariate time series is difficult to take advantage of spatial correlations, resulting in low detection accuracy. Furthermore, it is also difficult to meet the demand for real-time detection for univariate methods.

To efficiently and accurately localize anomalies in the entire metro system, we mainly focus on the anomaly passenger-flow detection for multivariate passenger flow data. In other words, the goal of our model is to find all anomalies at all stations simultaneously. For multi-stations passenger-flow anomaly detection task, there are two key issues that need to be addressed: 1) how to accurately identify anomalies based on spatio-temporal data; and 2) how to achieve real-time or early stage detection. For anomaly detection of multivariate data, the classic RPCA model is very suitable for the problem of abnormal passenger flow detection in entire metro system. However, the classic RPCA model ignores the unique spatio-temporal characteristics of regular passenger flow, which leads to a decrease in the accuracy of anomaly extraction. Inspired by classic RPCA, an innovative metro passenger flow anomaly detection approach, called real-time spatio-temporal RPCA model (RT-STRPCA), is proposed in this paper. The key idea in the proposed model is to decompose the observation data into two parts, i.e. normal component and abnormal component. The proposed model exploits a global consistency of metro passenger flow by using the low-rank nature of normal data and the sparsity of anomalies, as well as a local consistency with spatio-temporal side information. Besides, a sliding window method is adopted to achieve real-time detection or detection of early stage. The main contributions of this paper include:

- A novel real-time anomaly detection model (RT-STRPCA) based on sliding windows is proposed, which integrates both global and local consistency of passenger flow data. Experimental results demonstrate that the proposed model can achieve accurate anomaly detection.
- RT-STRPCA separates the abnormal component from normal component of passenger flow data by imposing a global low rank component on normal matrix and a sparsity regularizer on anomaly matrix. The low rank matrix represents the regular matrix which is generated by normal daily operations of metro system; the anomaly matrix represents anomaly matrix which is generated by unusual events.
- RT-STRPCA captures local consistency robustly according to the temporal periodicity and spatial similarity by posing temporal Toeplitz regularization and spatial Laplace regularization on regular matrix. The local consistency is integrated into the proposed model to improve the performance of anomaly detection based on passenger flow data.

The paper is organized as follows: the related research works are outlined in Section 2; The problem statement and

the real-time spatio-temporal RPCA-based model are given in Section 3; Section 4 conducts the experiments on Synthetic data and real passenger flow data to verify the feasibility and validity of RT-STRPCA; Section 5 concludes this paper with probable future work.

2 | RELATED WORKS

In this section, we review two relevant research fields including research works of anomaly detection in the transportation field and anomaly detection based on the matrix or tensor.

2.1 | Anomaly detection in the transportation field

In intelligent transportation systems, a large amount of various traffic data is accumulated. Mining traffic data to detect traffic anomalies has attracted extensive attention in recent years. Thus, there has been a large amount of research works on anomaly detection. A variety of different research interests are embedded in these research works. For example, Zhang et al. [1] considered the similar traffic dynamics of different urban regions as features and identified anomalous regions based on the similarity changing rate. Zhao et al. [2] used statistical-based methods to investigate individual travel patterns in terms of anomaly detection and regularity discovery. Zhang et al. [3] also detected abnormal taxi trajectories. Wang et al. [4] develop an anomaly detection framework focusing on identifying anomalous dynamics in traffic data. Pang et al. [5] adapted the likelihood ratio test statistic method to monitor anomalous traffic conditions. Dong et al. [6] used GPS trajectory data of vehicles to discover traffic jams. The research works [7] and [8] diagnosed traffic anomalies. The anomaly detection studied in this article focuses on discovering anomalous metro (traffic) states by detecting anomaly passenger (traffic) flow data.

In recent years, there have been many new works proposed for traffic flow data anomaly detection tasks. Some research works adapted feature-based methods to flag anomalies. The feature-based methods use features (average speed, distance etc.) from traffic datasets and then apply classical anomaly detection algorithms to identify anomalies. Wang et al. [9] used the average vehicle speed to detect traffic accidents or congestion. In addition, some works utilized time series decomposition models to detect anomalies. Zhang et al. [10] decomposed the traffic series naturally into three components: daily periodical component, weekly trend component and residual component. The residual component is considered as noises or unusual events. Besides, there are some works specifically for the abnormal detection of metro passenger flow. Gu et al. [11] developed a systematic approach combining a K-means algorithm and hierarchical clustering to identify anomalies at metro stations. Wang et al. [12] developed an elastic abnormality detection for out-flows to real-time detect the abnormal out-flows. Most of these existing works mainly consider the temporal characteristics of passenger (traffic) flow data, while ignoring the spatial

characteristics of the data. In metro networks, there are obvious spatial characteristics in passenger flow data. Therefore, most of them may just show limited ability due to ignoring spatial features.

To address the aforementioned research gaps, an anomaly detection method that considers both temporal and spatial characteristics is proposed to detect anomalies over time based on sliding windows.

2.2 | Anomaly detection based on the matrix or tensor

Spatio-temporal traffic state data can be organized as multivariate time series, for which matrix or tensor provides a natural representation. By representing traffic state data as a matrix or tensor, techniques such as principal component analysis (PCA) are used to process traffic state data. Some research works (refs. [13–15] etc.) that use PCA to the traffic anomaly detection have shown some promising initial results. The PCA-based methods are effective when the corruption is caused by small additive noise. The traditional PCA-based methods fail under the large corruption, even if the corruption affects only very few of the observations [16]. To achieve better data separation, Candès et al. [17] proposed Robust PCA (RPCA) which can decompose a matrix into a normal matrix and a sparse anomaly matrix. The RPCA-based approaches have been successfully applied in numerous areas such as background modelling [17], video restoration [18], image alignment [19], image recovery [20]. However, there are limited research works that use RPCA to detect anomalies in the transportation field. Besides, RPCA does not consider spatial and temporal side information. Therefore, directly applying RPCA to the metro system for anomaly detection has limited performance. With the deepening of the metro network, passengers' demands for travel efficiency and travel quality have become increasingly obvious, which means higher requirements for more sensitive and timely anomaly detection. In short, it is necessary extensively to investigate the temporal and spatial characteristics of passenger flow data in order to improve the performance of anomaly detection.

3 | METHODOLOGY

Figure 1 depicts the main procedure of anomaly detection task in offline state by using RT-STRPCA model. The left panel of Figure 1 illustrates that an observation matrix is organized from observed passenger flow data. The middle panel shows the implement of the proposed model and the right panel includes an anomaly matrix $\mathbf{S}$. The procedure of real-time anomaly detection is the same as that of offline anomaly detection. The difference is that real-time anomaly detection task requires input data to be updated in real-time through a sliding window. Next, the RT-STRPCA model is elaborated.

The proposed framework of RT-STRPCA is outlined in Figure 2, in which the passenger flow data is considered as a

matrix that consists current and historical passenger flow data. The proposed model decomposes the observation data into two parts, i.e. normal component and abnormal component.

- The normal component can be regarded as passenger flow data generated by regular daily operations. Typically, there are strong spatio-temporal dependencies in the passenger flow data, which means the regular (normal) passenger flow data has a low rank property. When the normal data is organized into a matrix, the matrix is low rank.
- The abnormal component can be considered as additional passenger flow data generated by abnormal events. Abnormal events are scarce and therefore abnormal values in passenger flow data are also scarce. When the abnormal component converts into a matrix, the matrix is sparse.
- As a kind of typical spatio-temporal data, passenger flow data has global consistency and local consistency. The low-rank nature of the regular matrix and the sparsity of the abnormal matrix can effectively capture the global consistency. Local consistency includes temporal periodicity and spatial similarity. To enhance the performance of the model in anomaly detection, the temporal Teoplitz regularization is introduced to encode periodic information and the spatial Laplacian constraint is used to impose similarity between two stations which share similar passenger flow pattern.
- Besides, to achieve real-time detection or detection of early stage, a sliding window method is integrated into the model. Ultimately, by solving RT-STRPCA, the original data matrix is decomposed into a low rank matrix and a sparse matrix. Through a sparse matrix, we can easily and quickly locate the stations where anomalies just started or are happening at the current time interval.

3.1 | Construction of passenger flow matrix

We denote by $\mathbf{X}$ the passenger flow data collected from $\mathbf{M}$ stations over $\mathbf{N}$ time intervals and x_{mn} the passenger flow value of station m at time interval n . Considering that passenger flow data accumulates over time, we set a sliding window on the time dimension, as illustrated in Figure 3. w is the size of the sliding window and it is set as a cycle time. The passenger flow matrix is updated as follows: The entries of the present time interval, which are denoted by red regions, are added. Meanwhile, the oldest entries denoted by blue regions are removed from the matrix.

Based on the passenger flow matrix, our principle concern is how to discover anomalies by utilizing implicit structures and internal relationships of passenger flow data.

3.2 | Metro network-level anomalies locating by RT-STRPCA model

Our aim is to detect anomalies in the passenger flow data based on the spatio-temporal matrix. The RPCA model is a powerful tool to discover anomalies by decomposing a given observation

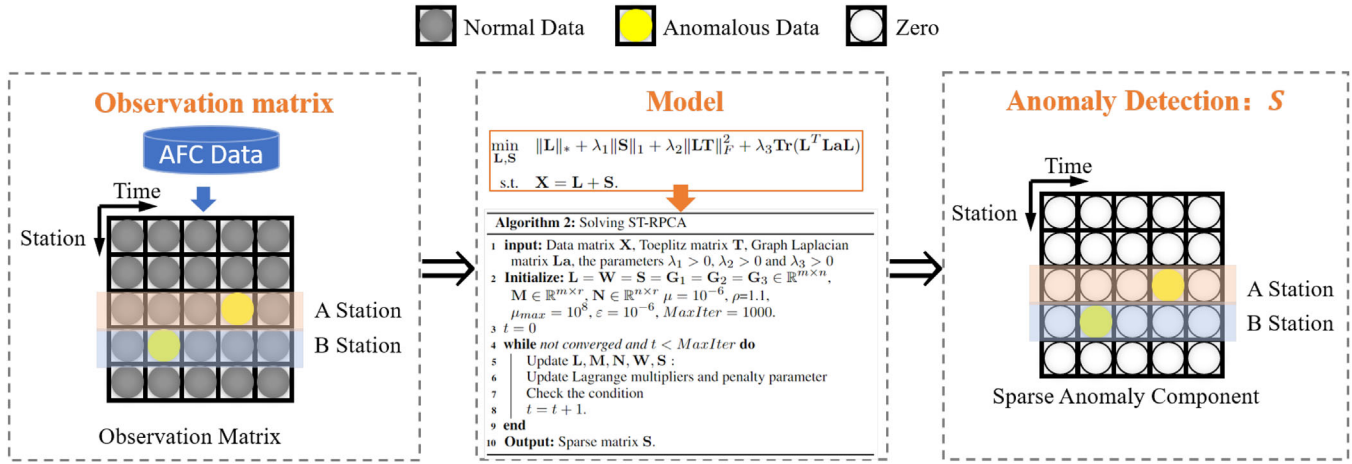


FIGURE 1 The main procedures for real-world data anomaly detection of offline state by using our model.

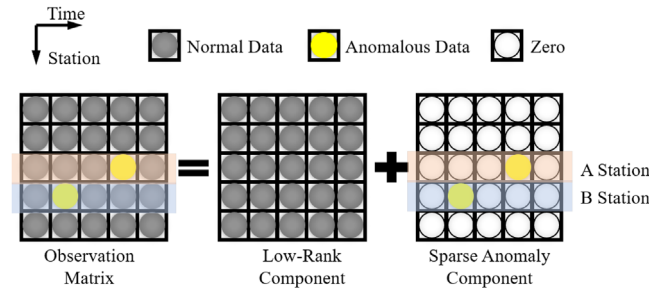


FIGURE 2 The framework of RT-STRPCA.

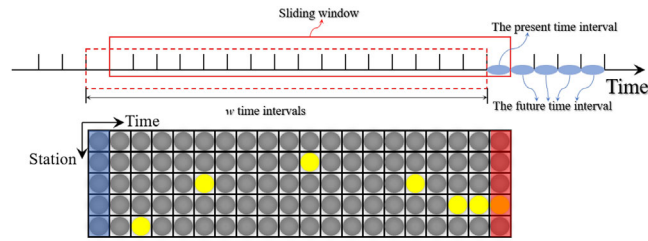


FIGURE 3 The sliding window and matrix update process.

matrix $\mathbf{X}$ into a low-rank component $\mathbf{L}$ and a sparse component $\mathbf{S}$, as follows:

$$\min_{\mathbf{L}, \mathbf{S}} \text{rank}(\mathbf{L}) + \lambda \|\mathbf{S}\|_0, \quad \text{s.t. } \mathbf{X} = \mathbf{L} + \mathbf{S}, \quad (1)$$

To make the problem solvable, the $\text{rank}(\mathbf{L})$ and the (ℓ_0) norm are replaced by their convex surrogates respectively (i.e. the nuclear norm and the ℓ_1 norm). The optimization problem (1) can be written as the following convex optimization problem:

$$\min_{\mathbf{L}, \mathbf{S}} \|\mathbf{L}\|_* + \lambda \|\mathbf{S}\|_1 \quad \text{s.t. } \mathbf{X} = \mathbf{L} + \mathbf{S}, \quad (2)$$

where λ is a positive weighting parameter. $\|\cdot\|_*$ is the nuclear norm. However, applying the RPCA model to detect passenger

flow data anomaly does not achieve satisfactory results, since it do not account for the spatio-temporal properties of the passenger flow matrix. Passenger flow data has been shown to have strong cyclical behaviour and obvious spatial structure. So we intend to incorporating side information into RPCA model to perform more accurate detection. To this end, we propose to solve the following optimization problem:

$$\begin{aligned} \min_{\mathbf{L}, \mathbf{S}} \quad & \|\mathbf{L}\|_* + \lambda_1 \|\mathbf{S}\|_1 + f(\mathbf{L}) + g(\mathbf{L}) \\ \text{s.t.} \quad & \mathbf{X} = \mathbf{L} + \mathbf{S}. \end{aligned} \quad (3)$$

where $f(\cdot)$ and $g(\cdot)$ are temporal and spatial constraints respectively which both act on low rank matrix $\mathbf{L}$. This model introduces robust temporal constraint by considering strong periodicity characteristic and incorporates spatial similarity operator to detect anomalies in passenger flow matrix.

3.2.1 | Temporal periodicity characteristics

In Figure 4, there is a straightforward observation. The passenger flow data shares the similar pattern on the same weekday or weekend of different weeks. In other words, the passenger flow data usually demonstrates a periodic pattern, which repeats with a certain frequency (such as 1 week). Thus the temporal information is introduced in the model and the optimization problem (3) is reformulated as below:

$$\begin{aligned} \min_{\mathbf{L}, \mathbf{S}} \quad & \|\mathbf{L}\|_* + \lambda_1 \|\mathbf{S}\|_1 + \lambda_2 \|\mathbf{L}\mathbf{T}\|_F^2 + g(\mathbf{L}) \\ \text{s.t.} \quad & \mathbf{X} = \mathbf{L} + \mathbf{S}. \end{aligned} \quad (4)$$

where the term $\|\mathbf{L}\mathbf{T}\|_F^2$ captures the temporal periodicity characteristics of the passenger flow matrix. The temporal constraint matrix $\mathbf{T} = \text{Toeplitz}(0, 1, -1)$, which denotes the Toeplitz matrix with central diagonal given by ones and the p th

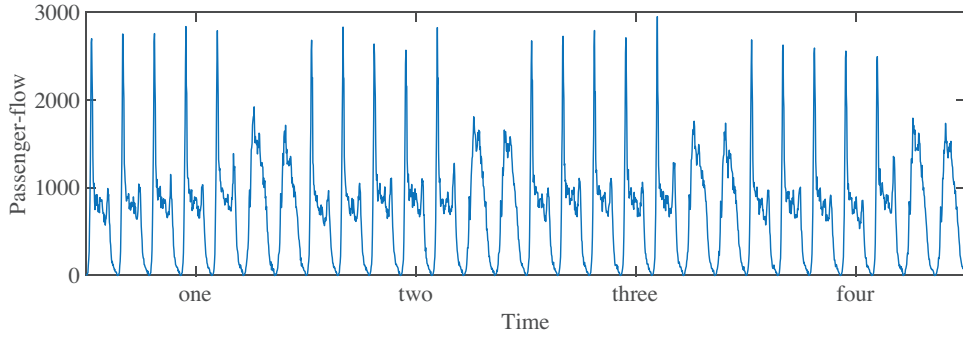


FIGURE 4 Temporal periodicity characteristics (the passenger flow exiting Xidan Station every 15 min for 1 month).

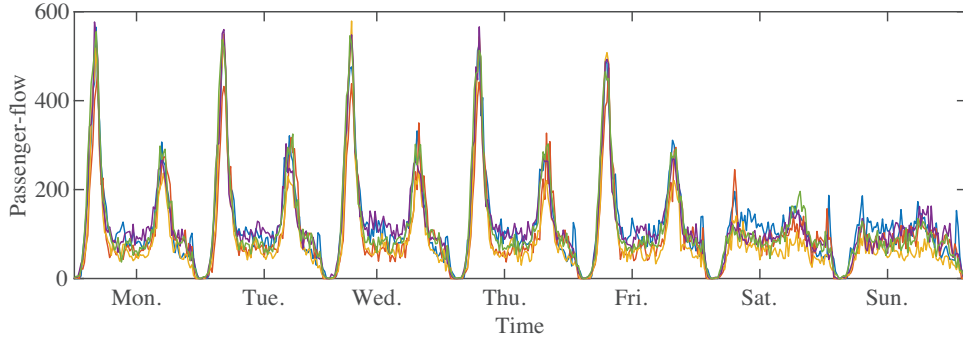


FIGURE 5 Spatial similarity characteristics (the passenger flow of several stations every 15 min for 1 week).

(p is the length of a cycle) lower diagonal given by negative ones, i.e.

$$\mathbf{T} = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & 0 & \cdots & 1 & 0 \\ 0 & -1 & \cdots & 0 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & -1 & 0 \\ 0 & 0 & \cdots & 0 & -1 \end{bmatrix}_{n \times (n-p)} \quad (5)$$

Thus, when left multiply $\mathbf{T}$ by $\mathbf{L}$, the difference of passenger-flow data at the same time between adjacent weeks can be realized by 1 and -1 . By minimizing $\|\mathbf{LT}\|_F^2$, we seek an approximation that also has the property of having similar value at the same time interval in the different weeks. The temporal constraint term intuitively expresses that there is usually similar passenger flow data in the same weekday or weekend of different weeks.

3.2.2 | Spatial similarity characteristics

There is local spatial similarity in passenger flow data of metro system, as shown in Figure 5. In this paper, the spatial correlation is defined as follows: two stations are considered spatially correlated if they have similar passenger flow patterns. But for spatial constraint, it's obvious inappropriate to apply a

simple form matrix like $\mathbf{T}$ for the following reasons. For the spatial similarity of metro system, it may lie not only in adjacent stations, but also in non-adjacent stations. These non-adjacent stations with spatial correlations usually share similar features, such as POI (point of interest) distribution. Moreover, the rows of the passenger flow matrix are ordered arbitrarily. Inspired by recent works, a graph regularizer is incorporated to our model to preserve the intrinsic spatial structure of the data. Specifically, let $G = (V, W)$ be the spatial graph, whose nodes V represent stations and weight W of edges represent similarity between stations. The similarity of the two stations are measured based on time series as follows:

$$\text{sim}(\ell_i, \ell_j) = \frac{1}{\|\ell_i - \ell_j\|_2} \quad (6)$$

where ℓ_i and ℓ_j is the time series of station i and station j , respectively. The threshold β is set to control the neighbourhood range and w_{ij} is given by:

$$w_{ij} = \begin{cases} \text{sim}(\ell_i, \ell_j), & \text{if } i \neq j \text{ and } \text{sim}(\ell_i, \ell_j) \geq \beta, \\ 0, & \text{else,} \end{cases} \quad (7)$$

The spatial constraint can be formulated as follows:

$$\begin{aligned} \frac{1}{2} \sum_{i,j=1}^N \|\ell_i - \ell_j\|_2^2 w_{ij} &= \text{Tr}(\mathbf{L}^T \mathbf{D} \mathbf{L}) - (\mathbf{L}^T \mathbf{W} \mathbf{L}) \\ &= \text{Tr}(\mathbf{L}^T \mathbf{L} \mathbf{a} \mathbf{L}) \end{aligned} \quad (8)$$

where $\text{Tr}(\cdot)$ stands for the trace of a matrix, $\mathbf{D}$ is a diagonal matrix in which $\mathbf{D}_{ii} = \sum_{j=1}^N \mathbf{W}_{ij}$, and $\mathbf{L}\mathbf{a} = \mathbf{D} - \mathbf{W}$ is the graph Laplacian matrix. Incorporating the graph regularization term into Equation (4) and the cost function are reformulated as follows:

$$\begin{aligned} \min_{\mathbf{L}, \mathbf{S}} \quad & \|\mathbf{L}\|_* + \lambda_1 \|\mathbf{S}\|_1 + \lambda_2 \|\mathbf{L}\mathbf{T}\|_F^2 + \lambda_3 \text{Tr}(\mathbf{L}^T \mathbf{L} \mathbf{a} \mathbf{L}) \\ \text{s.t.} \quad & \mathbf{X} = \mathbf{L} + \mathbf{S}. \end{aligned} \quad (9)$$

We call the model in Equation (9) spatial and temporal constrained RPCA (STRPCA) method. The sliding window is integrated into STRPCA model to achieve real-time anomaly detection and the method is called as real time STRPC (RT-STRPC) model. In order to solve the optimization problem in STRPCA, we adopt the alternating direction method of multipliers (ADMM) [21]. In each iteration of the optimization, it needs to compute SVD of matrix $\mathbf{L}$, which leads to a huge storage cost and computational overhead when solving our problem for big data. Hence, the nuclear norm for matrix $\mathbf{L}$ is substituted with its equivalent form defined by the below matrix factorization problem [22]:

$$\begin{aligned} \|\mathbf{L}\|_* = \min_{\mathbf{M} \in \mathbb{R}^{m \times r}, \mathbf{N} \in \mathbb{R}^{n \times r}} \quad & \frac{1}{2} (\|\mathbf{M}\|_F^2 + \|\mathbf{N}\|_F^2), \\ \text{s.t.} \quad & \mathbf{L} = \mathbf{M}\mathbf{N}^T, \end{aligned} \quad (10)$$

where r is the rank. Besides, we also introduce an auxiliary variable $\mathbf{L} = \mathbf{W}$, and Equation (9) is rewritten as below:

$$\begin{aligned} \min_{\mathbf{L}, \mathbf{M}, \mathbf{N}, \mathbf{S}, \mathbf{W}} \quad & \frac{1}{2} (\|\mathbf{M}\|_F^2 + \|\mathbf{N}\|_F^2) + \lambda_1 \|\mathbf{S}\|_1 + \lambda_2 \|\mathbf{L}\mathbf{T}\|_F^2 \\ & + \lambda_3 \text{Tr}(\mathbf{W}^T \mathbf{L} \mathbf{a} \mathbf{W}) \\ \text{s.t.} \quad & \mathbf{X} = \mathbf{L} + \mathbf{S}, \mathbf{L} = \mathbf{M}\mathbf{N}^T, \mathbf{L} = \mathbf{W}. \end{aligned} \quad (11)$$

Then we remove the linear equality constraints in Equation (9) by using the augmented Lagrangian method and have

$$\begin{aligned} \min_{\mathbf{L}, \mathbf{M}, \mathbf{N}, \mathbf{S}, \mathbf{W}} \quad & \frac{1}{2} (\|\mathbf{M}\|_F^2 + \|\mathbf{N}\|_F^2) + \lambda_1 \|\mathbf{S}\|_1 + \lambda_2 \|\mathbf{L}\mathbf{T}\|_F^2 \\ & + \lambda_3 \text{Tr}(\mathbf{W}^T \mathbf{L} \mathbf{a} \mathbf{W}) + \langle \mathbf{G}_1, \mathbf{X} - \mathbf{L} - \mathbf{S} \rangle \\ & + \langle \mathbf{G}_2, \mathbf{L} - \mathbf{M}\mathbf{N}^T \rangle + \langle \mathbf{G}_3, \mathbf{L} - \mathbf{W} \rangle \\ & + \frac{\mu}{2} (\|\mathbf{X} - \mathbf{L} - \mathbf{S}\|_F^2 + \|\mathbf{L} - \mathbf{M}\mathbf{N}^T\|_F^2 \\ & + \|\mathbf{L} - \mathbf{W}\|_F^2), \end{aligned} \quad (12)$$

with the definition $\langle \mathbf{B}, \mathbf{C} \rangle = \text{Tr}(\mathbf{B}^T \mathbf{C})$, $\mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3$ are Lagrangian multipliers, $\mu > 0$ is a positive penalty scalar.

For ST-RPCA model, $\mathbf{M}, \mathbf{N}, \mathbf{S}, \mathbf{W}, \mathbf{L}$ and other parameters can be solved by alternative iterations, in which superscript t denotes the current iteration step.

Updating variable $\mathbf{L}$: When $\mathbf{M}, \mathbf{N}, \mathbf{S}, \mathbf{W}$ are fixed, Equation (12) degenerates into a function with respect to $\mathbf{L}$. So we solve $\mathbf{L}$ by the following optimization problem:

$$\begin{aligned} \mathbf{L}^{(t+1)} = \arg \min_{\mathbf{L}} \quad & \lambda_2 \|\mathbf{L}^{(t)} \mathbf{T}\|_F^2 + \frac{\mu}{2} \left(\left\| \mathbf{X} - \mathbf{L}^{(t)} - \mathbf{S}^{(t)} + \frac{\mathbf{G}_1^{(t)}}{\mu^{(t)}} \right\|_F^2 \right. \\ & + \left\| \mathbf{L}^{(t)} - \mathbf{M}^{(t)} \mathbf{N}^{(t)T} + \frac{\mathbf{G}_2^{(t)}}{\mu^{(t)}} \right\|_F^2 \\ & \left. + \left\| \mathbf{L}^{(t)} - \mathbf{W}^{(t)} + \frac{\mathbf{G}_3^{(t)}}{\mu^{(t)}} \right\|_F^2 \right), \end{aligned} \quad (13)$$

Taking derivative of Equation (13) and set it to 0, the close-form solution is given by

$$\begin{aligned} \mathbf{L}^{(t+1)} = \mu^{(t)} (\mathbf{X} - \mathbf{S}^{(t)} + \frac{\mathbf{G}_1^{(t)}}{\mu^{(t)}} + \mathbf{M}^{(t)} \mathbf{N}^{(t)T} - \frac{\mathbf{G}_2^{(t)}}{\mu^{(t)}} \\ + \mathbf{W}^{(t)} - \frac{\mathbf{G}_3^{(t)}}{\mu^{(t)}}) (2\lambda_2 \mathbf{T} \mathbf{T}^T + 3\mu \mathbf{I}^n)^{-1} \end{aligned} \quad (14)$$

In a similar way with updating the variable $\mathbf{L}$, the close-form solutions of $\mathbf{M}, \mathbf{N}, \mathbf{W}$ are given by

$$\mathbf{M}^{(t+1)} = (\mu^{(t)} \mathbf{L}^{(t+1)} + \mathbf{G}_2^{(t)}) \mathbf{N}^{(t)} (\mathbf{I}^r + \mu \mathbf{N}^{(t)T} \mathbf{N}^{(t)})^{-1}, \quad (15)$$

$$\mathbf{N}^{(t+1)} = (\mu^{(t)} \mathbf{L}^{(t+1)} + \mathbf{G}_2^{(t)})^T \mathbf{M}^{(t+1)} (\mathbf{I}^r + \mu \mathbf{M}^{(t+1)T} \mathbf{M}^{(t+1)})^{-1}, \quad (16)$$

$$\mathbf{W}^{(t+1)} = (2\lambda_3 \mathbf{L} \mathbf{a} + \mu \mathbf{I}^m)^{-1} (\mu \mathbf{L}^{(t+1)} + \mathbf{G}_3^{(t)}), \quad (17)$$

Updating the variable $\mathbf{S}$: When others are fixed, Equation (12) degenerates into a function with respect to $\mathbf{S}$ as follows:

$$\begin{aligned} \mathbf{S}^{(t+1)} = \arg \min_{\mathbf{S}} \quad & \lambda_1 \|\mathbf{S}^{(t)}\|_1 \\ & + \frac{\mu^{(t)}}{2} \left\| \mathbf{X} - \mathbf{L}^{(t+1)} - \mathbf{S}^{(t)} + \frac{\mathbf{G}_1^{(t)}}{\mu^{(t)}} \right\|_F^2, \end{aligned} \quad (18)$$

The variable $\mathbf{S}$ is given by

$$\mathbf{S}^{(t+1)} = \text{sgn}(\mathbf{J}^{(t)}) \max(|\mathbf{J}^{(t)}| - \lambda_1 / \mu^{(t)}, 0), \quad (19)$$

where $\mathbf{J}^{(t)} = \mathbf{X} - \mathbf{L}^{(t+1)} + \frac{\mathbf{G}_1^{(t)}}{\mu^{(t)}}$.

ALGORITHM 1 Solving ST-RPCA.

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1 input: Data matrix  $\mathbf{X}$ , Toeplitz matrix  $\mathbf{T}$ , Graph Laplacian
   matrix  $\mathbf{L}_a$ , the parameters  $\lambda_1 > 0, \lambda_2 > 0$  and  $\lambda_3 > 0$ 
2 Initialize:  $\mathbf{L} = \mathbf{W} = \mathbf{S} = \mathbf{G}_1 = \mathbf{G}_2 = \mathbf{G}_3 \in \mathbb{R}^{m \times n}$ ,
    $\mathbf{M} \in \mathbb{R}^{m \times r}$ ,  $\mathbf{N} \in \mathbb{R}^{n \times r}$ ,  $\mu = 10^{-6}$ ,  $\rho = 1.1$ ,
    $\mu_{max} = 10^8$ ,  $\varepsilon = 10^{-6}$ ,  $MaxIter = 1000$ .
3  $t = 0$ 
4 while not converged and  $t < MaxIter$  do
5   Update  $\mathbf{L}$  : by equation(14)
6   Update  $\mathbf{M}$  : by equation(15)
7   Update  $\mathbf{N}$  : by equation(16)
8   Update  $\mathbf{W}$  : by equation(17)
9   Update  $\mathbf{S}$  : by equation(19)
10  Update Lagrange multipliers and penalty parameter: by
    equation(20)
11  Check the condition in equation(21)
12   $t = t + 1$ .
13 end
14 Output: Sparse matrix  $\mathbf{S}$ .

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Update $\mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3, \mu$: The Lagrangian multipliers $\mathbf{G}_1, \mathbf{G}_2, \mathbf{G}_3$, and penalty parameter μ are updated as follows:

$$\begin{aligned}
\mathbf{G}_1^{(t+1)} &= \mathbf{G}_1^{(t)} + \mu^{(t)} (\mathbf{X} - \mathbf{L}^{(t+1)} - \mathbf{S}^{(t+1)}) \\
\mathbf{G}_2^{(t+1)} &= \mathbf{G}_2^{(t)} + \mu^{(t)} (\mathbf{L}^{(t+1)} - \mathbf{M}^{(t+1)} \mathbf{N}^{(t+1)T}) \\
\mathbf{G}_3^{(t+1)} &= \mathbf{G}_3^{(t)} + \mu^{(t)} (\mathbf{L}^{(t+1)} - \mathbf{W}^{(t+1)}) \\
\mu^{(t+1)} &= \min(\rho \mu^{(t)}, \mu^{max}),
\end{aligned} \tag{20}$$

where $\rho > 1$ is a constant and μ^{max} is the upper bound of μ .

Convergence conditions: The stop criterion is measured as follows:

$$\begin{aligned}
\|\mathbf{X}^{(t+1)} - \mathbf{L}^{(t+1)} - \mathbf{S}^{(t+1)}\|_\infty &\leq \varepsilon, \\
\|\mathbf{L}^{(t+1)} - \mathbf{M}^{(t+1)} \mathbf{N}^{(t+1)T}\|_\infty &\leq \varepsilon, \\
\|\mathbf{L}^{(t+1)} - \mathbf{W}^{(t+1)}\|_\infty \leq \varepsilon, \|\mathbf{L}^{(t+1)} - \mathbf{L}^{(t)}\|_\infty &\leq \varepsilon, \\
\|\mathbf{S}^{(t+1)} - \mathbf{S}^{(t)}\|_\infty \leq \varepsilon, \|\mathbf{M}^{(t+1)} - \mathbf{M}^{(t)}\|_\infty &\leq \varepsilon, \\
\|\mathbf{N}^{(t+1)} - \mathbf{N}^{(t)}\|_\infty \leq \varepsilon, \|\mathbf{W}^{(t+1)} - \mathbf{W}^{(t)}\|_\infty &\leq \varepsilon,
\end{aligned} \tag{21}$$

where ε is tolerance error. If the convergence conditions is met, the iteration terminates.

The overall algorithm is summarized in Algorithm. 1.

3.3 | Anomaly detection

In general, normal passenger flow patterns lie in a low dimensional subspace despite the high dimensionality of collected data. The RT-STRPCA can separate the low-dimensional normal subspace from the abnormal subspace. Through projecting onto the abnormal subspace, anomalies can be easily identified. In passenger flow data, the anomaly threshold and the impact of anomalies might vary a lot to distinct stations. For example, the average entry flow for a metro station A is 100 at 7 am, while

for station B is 1000. As for the same 100 population increase, 200 entry flow of A at 7 am can be regarded as anomaly, but it is hard to say 1100 is also abnormal in station B. In this case, it is inappropriate to use the same threshold to detect anomalies for all stations. Therefore, we determine these thresholds for all stations in a simple way:

$$\delta_{mm} = cL_{mm}, \tag{22}$$

where L_{mm} denotes ideal passenger flow data of station m at the n th time interval; c is a constant that has an important impact on the selection of candidate anomalies.

4 | EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we conduct experiments on synthetic and real world traffic data to demonstrate the superior performance of our proposed method over the state-of-the-art alternative methods.

4.1 | Experimental datasets

We will carry out experiments by using synthetic and real world datasets. Simulations are necessary for accurate comparisons of anomaly detection task and Ringberg et al. [23] explained this in detail. In brief, they explained that: (1) accurate and complete ground truth information is needed to calculate evaluation metrics; (2) many more results in different anomalous conditions are needed to form accurate estimates of performance, and (3) simulations contribute to various parameters (it refers to the anomaly rate in this paper) that comprehensively present the detection ability of models. Simulation is necessary, but not sufficient for validation, so we also will conduct experiments on real world datasets.

(1) Simulated datasets for anomaly detection:

Simulations are generated based on real world passenger flow data which is from AFC in metro system provided by Beijing Transportation Information Center. Simulations are not completely realistic, but it can clearly illustrate the properties of different anomaly detection methods. Simulations are generated in three steps:

- The normal passenger flow data are generated. Since the collected data unavoidably contains small dense additive noise and a few anomalies, which may conflicts with the constructed anomalies, the original data is denoised. The normal datasets still retain local spatio-temporal structure the same as the real world datasets.
- Noises are added by multiplying 0.1 times the normal passenger flow with a factor sampled from a normal distribution with mean 0.25 and variance 0.04.
- Anomalies are injected into the datasets with noises. When assuming that there is an anomalous point, the size of the

anomaly is set to 0.5 times average value of dataset with noises. The anomaly rate varies from 1% to 35%.

In this paper, we set up three typical anomaly scenarios: random anomaly, temporal-mode anomaly and spatial-mode anomaly, as illustrated in Figure 6.

- Random-mode anomaly: anomaly elements are randomly chosen, as presented in Figure 6(a).
- Temporal-mode anomaly: we randomly select 50% rows and continuous elements with a certain rate in each row are chosen, which is shown in as presented in Figure 6(b).
- Spatial-mode anomaly: we randomly select 50% columns and continuous elements with a certain rate in each column are chosen, as presented in Figure 6(c–e).

All of the matrix-based models are applied to acquire sparse matrix and detect anomalies by setting threshold on sparse matrix.

(2) Real-world datasets:

In addition to the performance evaluation of the model via synthetic datasets, we also use two real-world datasets to verify its applicability in real-time anomaly detection. The real world event detection experiment is conducted on the AFC datasets collected from Beijing Metro. One dataset is out-flow data of metro network from 5 days (23 Jul, 30 Jul, 6 Aug, 13 Aug, 20 Aug, respectively) in 2015. The other dataset is in-flow data of metro network from 5 days (9 Aug, 16 Aug, 23 Aug, 30 Aug, 6 Sep, respectively) in 2015. The size of the time sliding window is set to 4 days. Meanwhile, we collect events in Sina Weibo with time and location to verify anomalies detected by our model. For example, {Qixi Festival, 23 Aug, 2015, Xidan} is the description, time, and location of one recorded event.

4.2 | Experimental setting

(1) Baseline models

We compare our model with three baselines, including one classic model: IForest (Isolation forest) [24], two matrix-based models: RPCA (described in (2)) and NMF (nonnegative matrix factorization) [25]. IForest is used to perform anomaly passenger-flow detection for each metro station at a time.

(2) Evaluation metric

Anomalies detected by all of models are compared with the injected anomalies to measure the detection performance of models. F_1 score is used as a metric, defined by:

$$\begin{aligned} F_1 &= 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}, \\ \text{Precision} &= \frac{\text{TP}}{\text{TP} + \text{FP}}, \\ \text{Recall} &= \frac{\text{TP}}{\text{TP} + \text{FN}}, \end{aligned} \quad (23)$$

where TP, FP, and FN are defined as the number of true positive instances, false positive instances and false negative instances respectively. The precision represents the proportion of true anomalies in all detected anomalies. The recall represents the proportion of correctly detected anomalies among all true anomalies.

(3) Parameters setting

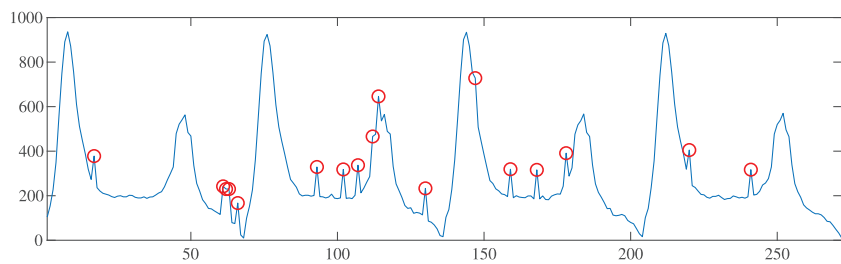
There are several parameters, i.e. the latent rank r , the threshold β , λ_1 , λ_2 , and λ_3 . In the following, we select these parameters based on the variability of STRPCA detection performance with different parameters values.

- Threshold β : The threshold controls the number of similar neighbours in spatial relations. As shown in Figure 7, the F_1 score increases for all datasets with β increasing. After $\beta = 0.002$, the F_1 score preserves relatively stable. Thus, we set $\beta = 0.002$ for all datasets.
- Latent rank r : The latent rank is often significantly lower than the dimensionality of the data [26], which reflects global low rank structure of data. In Figure 8, we visualize the variations in our model's detection performance caused by different ranks on datasets under different abnormal rates (from 1% to 35%). It can be found that the F_1 score of most datasets is relatively higher within the lower latent rank range. Considering the computation load caused by the rank, we choose 15 as input latent rank in our experiments for all datasets and all methods.
- Regularization parameters λ_1 , λ_2 , and λ_3 : λ_1 , λ_2 , and λ_3 are non-negative weights corresponding to sparse constraint term, temporal constraint term and spatial constraint term and they are tuned based on experiments. Following ref. [27], the values of these parameters are selected from 10^{-3} to 10^3 by using cycle procedure and the detection performance on datasets by varying these parameters is illustrated in Figure 9. Finally, these parameters are determined for all datasets as follows: $\lambda_1 = 1$, $\lambda_2 = 1$, and $\lambda_3 = 0.1$.

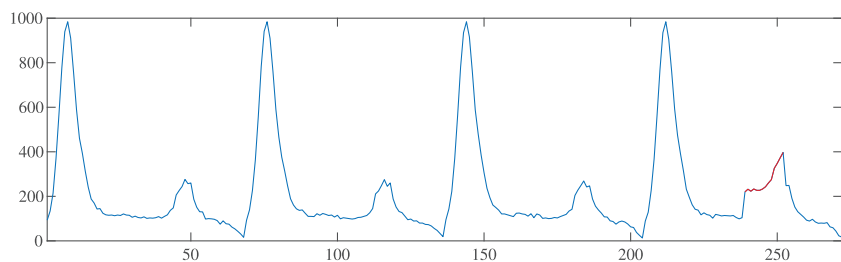
4.3 | Experimental results analysis on synthetic datasets

The dataset for offline state contain 4-weeks data, while the dataset for real-time state consist 5-weeks data and the size of sliding window is 4 weeks. In offline detection, we compared the STRPCA model with three other models: IForest, RPCA, NMF. In real-time detection, we compared the RT-STRPCA model with RPCA model.

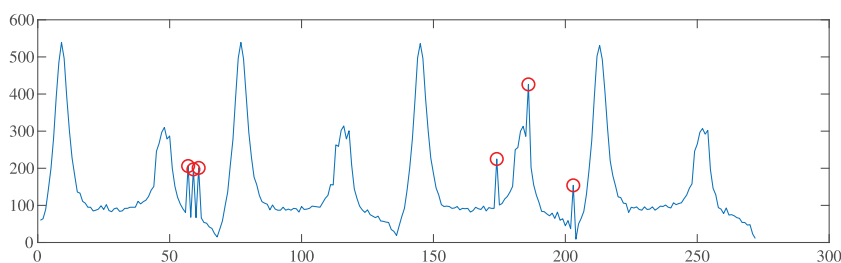
Figure 10 visualizes the comparison results of STRPCA with the baseline models in an offline state. Intuitively, it can be seen that the detection performance of our model significantly outperforms the baseline model in all abnormal modes. As the anomaly rate increases, the detection performance of models (except for IForest) show a downward trend. Figure 10(a) illustrates the performance comparison in random anomaly mode. Compared with other two modes (Figure 10(b,c)), the



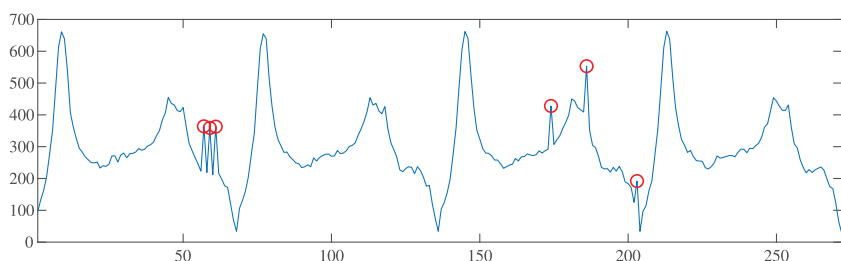
(a) Random Anomaly



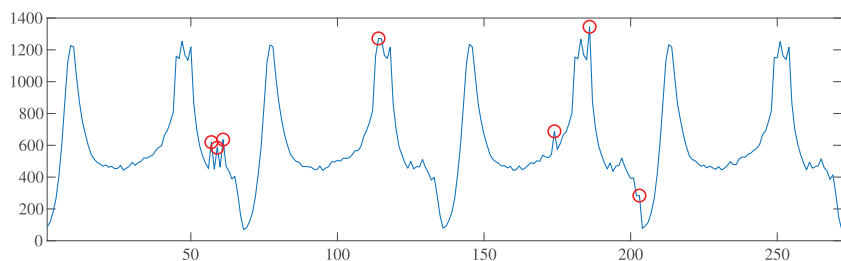
(b) Temporal-mode Anomaly



(c) Spatial-mode Anomaly(station 94)



(d) Spatial-mode Anomaly(station 95)



(e) Spatial-mode Anomaly(station 96)

FIGURE 6 Synthetic datasets containing multiple types of anomalies.

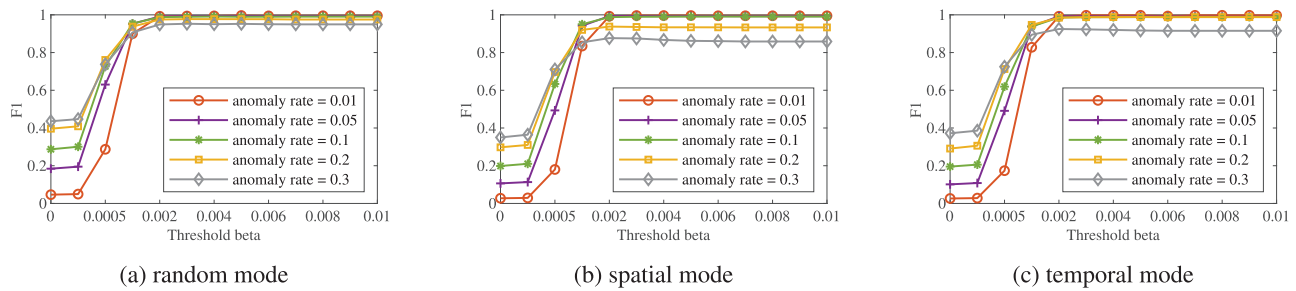


FIGURE 7 F_1 score of STRPCA by varying threshold β for different anomaly rate.

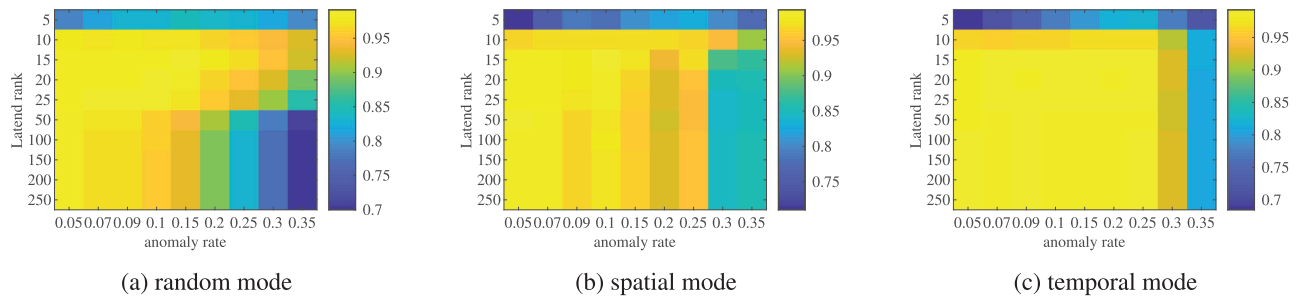


FIGURE 8 F_1 score of STRPCA by varying latent rank r for different anomaly rate.

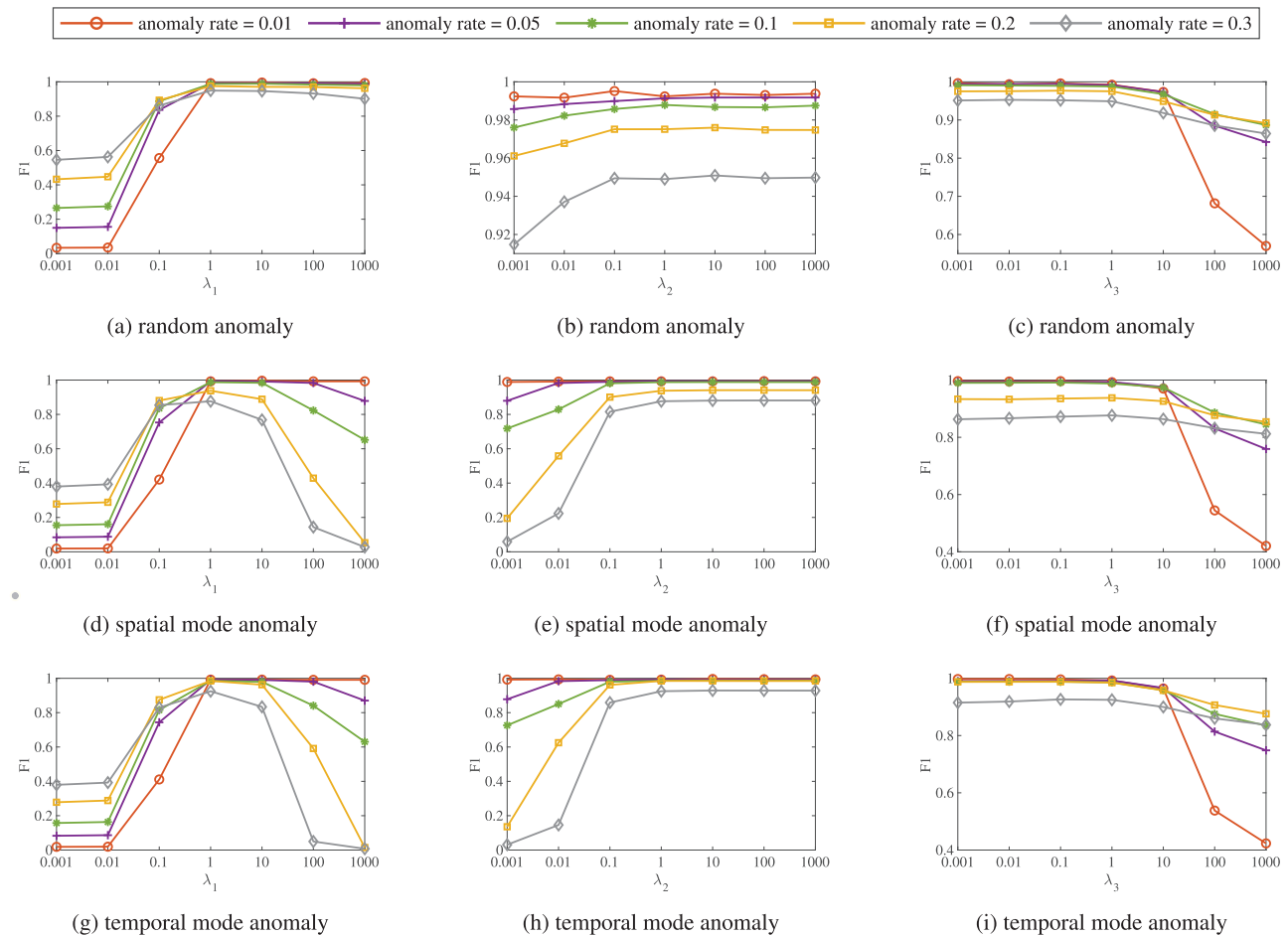


FIGURE 9 F_1 score of STRPCA by varying parameter λ_1 , λ_2 , and λ_3 for different anomaly rate.

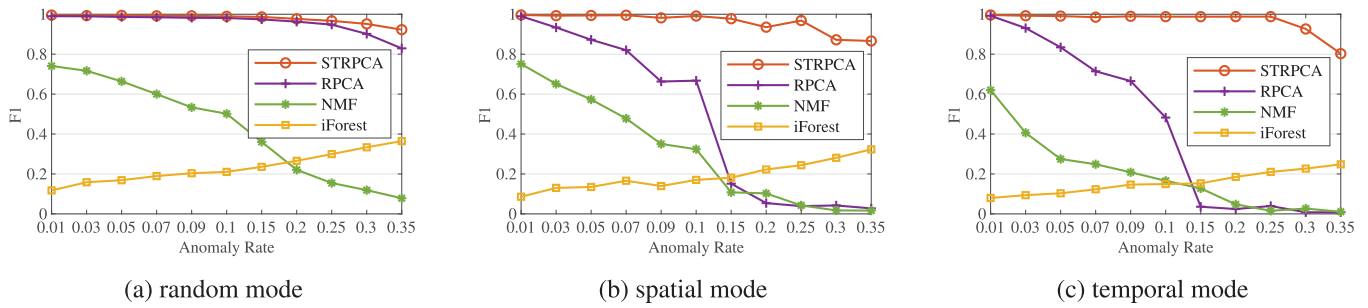


FIGURE 10 Performance comparison of different models in offline state.

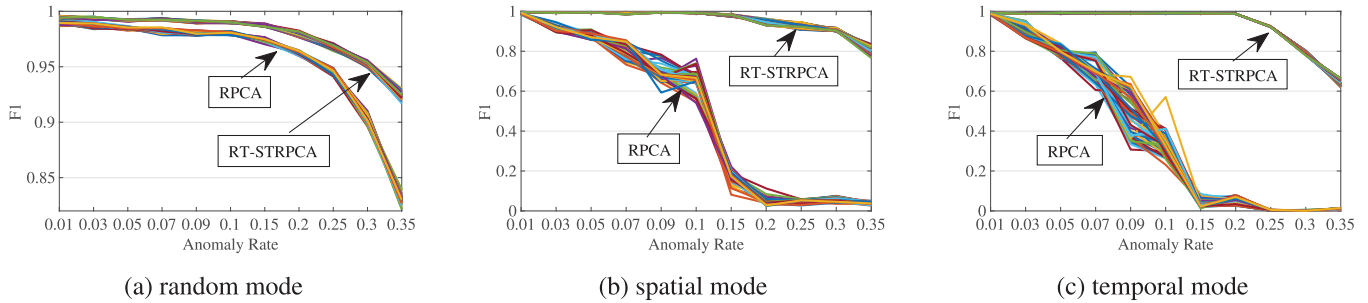


FIGURE 11 Performance comparison of different models in real time.

performance of RPCA and STRPCA has a gradual decrease. In other two modes, the detection performance of RPCA model decreases significantly while our model still maintains stable detection performance and F_1 score is above 80%. The above results fully demonstrate that auxiliary information is significantly helpful to improve the robustness of anomaly detection. The real time experimental results are shown in Figure 11. For each time step we move, we calculate an F_1 score. The sliding window has been slid 68 times in total and all the results are shown in Figure 11. Clearly, the detection performance within 68 windows is almost the same and the performance of online detection is basically the same as that of offline detection. For all datasets, RT-STRPCA model is much better than the baseline model.

In conclusion, our method outperforms the baseline methods on synthetic datasets. The results show that our method has strong robustness in anomaly detection and the spatio-temporal information extracted by our method improves the anomaly detection performance.

4.4 | Experimental results analysis on real-world datasets

Figures 12 and 13 demonstrate the detection results of the RT-STRPCA model on two real-world datasets. For out-flow dataset, taking Xidan Station as an example, Figure 12 visualized the anomalies identified in the real-time detection process. It is not difficult to see from Figure 12(a) that our model easily detected the appearance of the anomaly. As the sliding win-

dow moves to the right, the model successfully captures the entire process of anomaly sequence change at the Xidan Station. To verify the anomaly sequence detected on Xidan Station, we collected information from Sina Weibo which is an influential social media and finally found events related to anomalies, as shown in Figure 14(a). 20 Aug, 2015 is Qixi Festival which is the most romantic one among the Chinese traditional festivals. The Qixi Festival is also known as the Chinese Valentine's Day. On the day, many shopping malls near Xidan station launched promotional activities, attracting a large number of people to go shopping and entertainment. Meanwhile, 20 Aug, 2015 is a working day, thus most people choose to go to the mall after getting off work in the evening. This phenomenon coincides with the changes of anomaly sequence detected at Xidan Station. In Figure 12(d), compared to other days, the passenger flow of the whole day (20 Aug, 2015) showed a rising change and the passenger flow increased significantly especially in the evening. For in-flow dataset, Figure 13 illustrates the real-time detection results of Sihui station. As shown in Figure 13(d), there are two anomaly temporal sequences at Sihui station. Judging from the event information searched from Sina Weibo, there were exactly two events affecting the passenger flow of Sihui Station. These two events occurred on 23 Aug, 2015 and 6 Sep, 2015 respectively, as shown in Figure 14(b,c). On 23 Aug, 2015, trains were out of operation from 0:00 am to 12:00 pm. According to Figure 13, the real-world passenger flow is 0 in the first half of 23 Aug, 2015. Although C is Sunday, people need to go to work. Thus, the passenger flow pattern at Sihui Station on 6 Sep, 2015 is completely different from that of other Sundays, as shown in Figure 13(d).

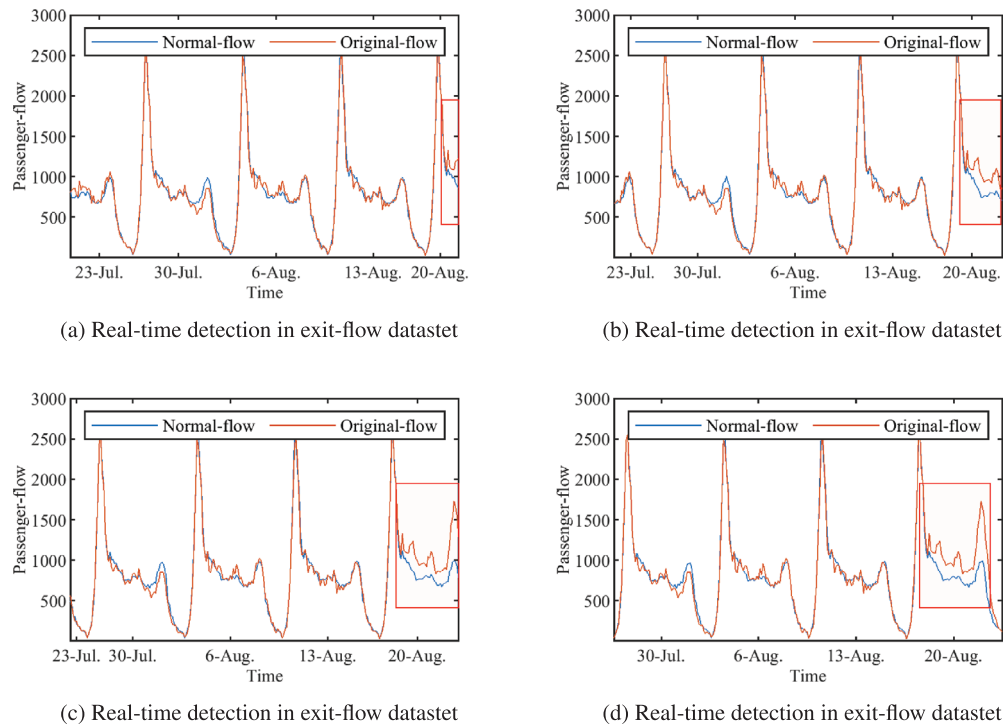


FIGURE 12 Real-time anomaly detection performance for different anomaly rate in two real-world datasets.

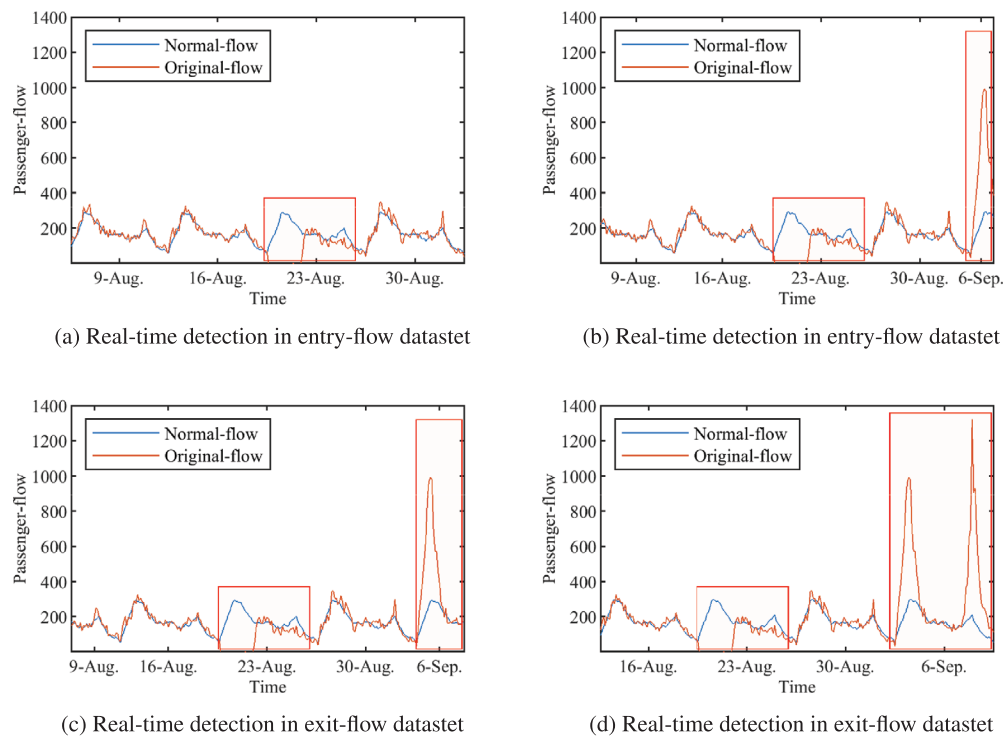
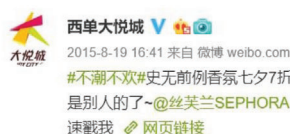
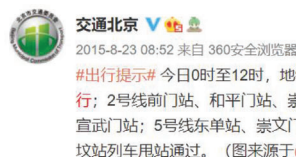


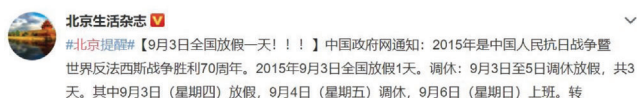
FIGURE 13 Real-time anomaly detection performance for different anomaly rate in two real-world datasets.



(a) Event 1: 20 Aug, 2015 is Qixi Festival which is the most romantic one among the Chinese traditional festivals. On the day, many shopping malls near Xidan station launched promotional activities, attracting a large number of people to go shopping and entertainment.



(b) Event 2: On 23 Aug, 2015, trains were out of operation from 0:00 am to 12:00 pm.



(c) Event 3: 6 Sep, 2015 is Sunday but exchanged with the working day because of legal exchanging holiday

FIGURE 14 Event.

In short, it is proved that our method has the ability to reliably detect anomalies in real time by the experimental results on real datasets and verification of social data, no matter the anomaly just started to happen or is continuing.

5 | CONCLUSIONS

In this paper, we propose a novel method called RT-STRPCA to real-time detect anomalies based on passenger flow data. Our method is an unsupervised method which has the ability to effectively detect abnormal passenger flow in real time, only taking a certain amount of historical and real-time flow data as input. Our model decomposes a passenger flow matrix into a low rank matrix and a sparse matrix. By the sparse matrix, anomalies can be identified. Passenger flow data is typically spatio-temporal data, which demonstrates both global consistency and local consistency. In RT-STRPCA model, the low rank matrix is effective to capture the global consistency. Meanwhile, local consistency is also integrated into the optimization framework to better characterize side information. The local consistency includes temporal periodicity and spatial similarity. Applied to anomaly passenger flow detection, our method outperforms other commonly used methods.

This work verifies the remarkable ability of RT-STRPCA model for low-rank matrices and sparse anomalies separation. Thus in the future, RT-STRPCA can be used to analyze other spatio-temporal data. Moreover, the domain knowledge of the spatio-temporal data can be taken into account to substitute the

spatial and/or temporal structure in the RT-STRPCA model for some improvement of results. Also, it is suggested to employ RT-STRPCA model for other applications, e.g. traffic data completion, traffic estimation and traffic prediction due to its effective spatio-temporal regularization.

AUTHOR CONTRIBUTIONS

Xiulan Wei: Investigation, methodology, validation, writing - original draft, writing - review and editing. Yong Zhang: Project administration, resources, supervision, writing - review and editing. Xinyu Zhang: Validation, writing - review and editing. Qibin Ge: Supervision, writing - review and editing. Baocai Yin: Resources, writing - review and editing.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

ORCID

Xiulan Wei <https://orcid.org/0000-0001-9237-3944>

Xinyu Zhang <https://orcid.org/0000-0001-6400-9475>

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